

Technical overview (Controller)

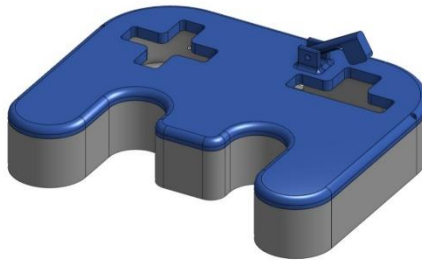


Figure 1: Controller design (before fine tuning!)

The architecture and communication

The controller is powered by an Arduino m0 controller, which is based on the cortex m0+ architecture. This microcontroller provides the processing power to handle multiple sensors simultaneously, ensuring a good communication.

The controller consists of various sensors that interact through digital pins, serial communication and I2C protocols, coordinated by a central microcontroller.

Communication overview:

- Digital inputs/outputs: for simple on/off interactions
- I2C bus: for communication with more complex sensors and displays
- Serial communication: For transmitting data and receiving commands, integrated with the HID-project library for keyboard 'emulation'.

The components

The input sensors:

Component	Controller Function	Communication Method
Buttons	Strafing and map switching	Digital input
Touch Sensor	Continuous movement forward	Digital input
Proximity Sensor (VCNL4040)	Weapon use (attack action)	I2C, HID keyboard interrupt
Reed relais	Menu navigation, fish cutting	Digital input (magnetic field detection)
IMU (BMI323)	View control / menu navigation	I2C, reading, UART serial communication

The output sensors:

Component	Controller Function	Communication Method
7-Segment-Displays	Displaying xp	I2C
Vibrator Motor	Feedback when picking up items	Digital output
Led array	Visual effect in chopping minigame	I2C

Controller Design

The controller features a mechanical design with a faceplate that includes cutouts for the sensor integrations and segment displays. In Figure 1 the cutouts for the led array and 7segment displays aren't made yet. We're currently still busy fine tuning the design. The IMU is embedded in the middle on that "weird" part between the two hand holders. A magnet is also concealed within the lever for the implementation of the reed relais. The overall structure is assembled using standard screw holes, allowing for easy assembly and potential upgrades.

Next Steps

Game <-> Minigame navigation: We are currently working on developing a motion based navigation, where the controller movement emulates mouse movement, similar to the Wii's motion controls. This feature is still under development.

It could also be interesting to look at safety nets such as a connection loss indicator, fault indicator to make the controller more user friendly.

SWOT analysis Fish to Dish Controller

Strengths: Originality of the game and idea Unique design (not creative) Innovative input methods	Weaknesses: Difficult user experience Time loss technical issues
Opportunities: More time= More creativity in design Process feedback	Threats: Users quit because of frustration Time limit

During our project, we developed an original game that is controlled through a custom-built controller. One of the biggest strengths of this project is the originality of the game concept. The idea itself is unique and stands out from traditional gameplay experiences. Although the controller's design is relatively simple and not particularly creative in its form, it does have an add on, the little lever on the controller is our additional asset. The technical implementation was successful overall. All the components and sensors function properly, including more unconventional elements like a reed switch, used to detect magnetic changes, and an IMU to track movement. The use of these less common can also be seen as a strength.

However, throughout the development process, we faced a number of challenges and weaknesses. The most significant issue was the difficulty of the user experience. Although everything works as intended, the controller feels a little unintuitive and somewhat hard to use, especially for someone who hasn't seen it before. This could negatively impact the overall enjoyment of the game. On top of that, we experienced a considerable loss of time due to technical issues, particularly related to the IMU. Since we couldn't rely on pre-existing libraries, we had to write our own, and to make things more difficult, the pinout diagram we found online was incorrect. As a result, we lost several valuable time troubleshooting these problems, which took time away from optimizing the usability and final product.

Despite these struggles, there are several opportunities. With more development time, we could have refined both the controller's physical design and the sensors logic to better match the creativity and uniqueness of the game itself. There is also great potential in gathering user feedback, testing the controller in real scenarios, and improving the user interface based on real input. This would help close the gap between the idea and its practical implementation.

Finally, we also have to acknowledge some threats to our project. Due to the difficulty of the controls, there is a risk that users might quickly lose interest or become frustrated, which could harm their overall impression of the game. Also the strict time constraints limited our ability to fully execute our ideas and polish the final result as much as we would have liked. These factors make it clear that while the foundation is solid, the controller and experience as a whole would benefit greatly from additional time.

Contributions (everyone wrote their own contributions)

Contributions Kayen Smet

- All Powerpoints, Swot, Technical overview ..
- Soldering Buttons
- Thought process of imu
- Proximity sensor code + implementation in game
- Touch sensor: worked together with Milo for code and implementation
- 7-segment displays: worked together with milo in the first few introduction classes after the basic tasks
- Hall effect sensor: tried implementing but the sensor was faulty
- Reed relais: worked with milo for code and implementation
- Soldering the reed relais: worked with Artuur
- Suggestions for controller design
- Changes in code for the game

Contributions Milo Regnard

- 3D design of the controller, printed the design
- Made the BMI 323 (IMU) work on the controller and work in game
- Wrote the code to make the controller work with the game (with Kayen)
- Mouse emulation
- Led array and vibrators
- Xp-display on 7segment displays
- Support team, find solutions,...

Contributions Artuur Mouton

- Helped with mapping and pinout of the IMU module
- Made documentation on the GPIO of IMU
- Pinout and connection of reed relais (helped Kayen)
- Soldering the reed relais on pcb
- Instagram video